Exhibit 10.1

TRANSITION AND SEPARATION AGREEMENT

This Transition and Separation Agreement ("Agreement") is made and entered into effective as of September 15, 2022, by and between J. Melville Engle ("Employee") and Predictive Oncology Inc., a Delaware corporation ("the Company"), each of whom enter into this Agreement intending to be legally bound.

RECITALS

The Company and Employee agree upon the following background facts and incorporate them by reference into this Agreement.

1. Employee and the Company entered into an Employment Agreement, effective March 19, 2021 (the "Employment Agreement").

2. The Company adopted a 2021 Long-Term Incentive Plan, effective May 17, 2021 (the "LTIP"), pursuant to which Employee was granted a restricted stock unit award.

3. Employee has decided to resign as an officer, employee and member of the Company's Board of Directors (the "Board") as of October 31, 2022 in connection with his retirement; and

4. Employee and the Company wish to enter into this Agreement to set forth the terms and conditions under which Employee will transition out of his current roles with the Company and voluntarily separate from his employment and Board position.

AGREEMENT

NOW, THEREFORE, based upon the foregoing recitals and for good and valuable consideration described below, the receipt and sufficiency of which are hereby expressly acknowledged, Employee and the Company agree as follows:

1. **Voluntary Resignation from Employment.** Employee will voluntarily resign his employment with the Company effective October 31, 2022 (the "Separation Date").

2. **Voluntary Resignation from Board of Directors Position**. Employee will voluntarily resign his position on the Company's Board of Directors effective October 31, 2022.

3. **Cooperation and Transition.** Employee understands that the Company's obligations under this Agreement, including all consideration set forth in this Agreement, are contingent upon, among other things, Employee cooperating with the Company in his transition which includes:

   (a) Employee will continue his employment pursuant to the terms of the Employment Agreement through October 31, 2022.

   (b) After October 31, 2022, Employee will make himself available, on a reasonable basis and during normal business hours, to discuss or address issues or questions relating to Employee's employment and position for the purpose of achieving a smooth transition of Employee's former job duties and responsibilities.

(c) As requested by the Company, Employee will execute all documentation necessary to remove himself as of October 31, 2022 from all positions he holds with the Company and all Company-related entities, including his director position.

(d) Employee also agrees to be available to cooperate with the Company and its counsel in connection with any investigation, administrative proceeding or litigation relating to any matter, occurring during Employee's employment, in which he was involved or of which he has knowledge. Employee understands and agrees that such cooperation includes, but is not limited to, making himself available to the Company and/or its counsel upon reasonable notice for: interviews and factual investigations; appearing to give testimony without requiring service of a subpoena or other legal process; volunteering to the Company or its counsel pertinent information; and turning over all relevant documents which are or may come into his possession.

4. **Separation Benefits.** In accordance with and subject to the terms of the Employment Agreement and the LTIP, Employee is entitled to certain separation benefits upon termination without cause. Notwithstanding that Employee is voluntarily resigning his employment and will not be terminated without cause, the Company will agree to provide Employee with the separation benefits in accordance with the Employment Agreement and the LTIP, as set forth in the Agreement and Release attached hereto as Exhibit 1. Employee understands that he needs to execute the Agreement and Release attached hereto as Exhibit 1 in order to receive these separation benefits.

5. **Non-Disparagement**. Employee agrees not to make disparaging or defamatory remarks about the Company or the Company's services, products, or other matters pertaining to its business. The Company, on behalf of itself and its management team, agrees not to make disparaging or defamatory remarks about Employee or Employee's services or other matters pertaining to Employee. Nothing in this paragraph is intended to, nor may be interpreted to, prevent the Employee or Company from giving truthful testimony to any law enforcement officer, court, administrative proceeding or as part of a government investigation.

6. **Additional Agreements and Understandings.**

   a. **Successors and Assigns.** This Agreement is personal to Employee and may not be assigned by Employee without the written agreement of the Company. The rights and obligations of this Agreement shall inure to the successors and assigns of the Company.

   b. **Severability.** If a court finds any term of this Agreement to be invalid, unenforceable, or void, Employee and the Company agree that the court shall modify such term to make it enforceable to the maximum extent possible. If the term cannot be modified, Employee and the Company agree that the term shall be severed and all other terms of this Agreement shall remain in effect.

c. **Entire Agreement.** This Agreement constitutes the sole understanding of Employee and the Company with respect to Employee's resignation from employment and from the Board. This Agreement may not be modified, altered, or changed in any way except by written agreement signed by Employee and an authorized representative of the Company.

d. **Other Agreements.** Nothing contained in this Agreement shall affect Employee's obligations owed to the Company under the Employment Agreement or any other agreement between Employee and the Company.

e. **No Waiver.** No claim or right arising out of a breach or default under this Agreement may be discharged by a waiver of that claim or right unless the waiver is made in writing and signed by an authorized representative of the Company. A waiver by any party of a breach or default of the other party of any provision contained in this Agreement shall not be deemed a waiver of future compliance of such provisions, and such provisions shall remain in full force and effect.

f. **Taxes.** Employee acknowledges that Employee has not relied on any tax advice provided by the Company and that, if necessary, Employee is solely responsible for properly reporting the tax consequences and paying any applicable taxes, penalties, and interest. Employee acknowledges and agrees that Employee has been provided with the opportunity to consult legal and financial counsel with respect to the tax treatment of the separation benefits provided in the Agreement and Release attached hereto as Exhibit 1. Employee has been advised by the Company to consult with such counsel.

g. **Governing Law/Venue.** The laws of the State of Minnesota will govern the validity, construction, and performance of this Agreement, without regard to the conflict of law provisions of any other jurisdictions. Employee irrevocably consents to the exclusive jurisdiction of courts in Minnesota for the purposes of any action arising out of or related to this Agreement or any dispute between the Company and Employee, including any actions for temporary, preliminary, and permanent equitable relief. Employee irrevocably waives Employee's right, if any, to have any disputes between the Company and Employee arising out of or related to this Agreement decided in any jurisdiction or venue other than a state or federal court in the State of Minnesota.

[*Signature page follows*]

IN WITNESS WHEREOF, Employee and the Company have executed this Agreement as of the date and year set forth below.

Dated: 9/15/2022

/s/ J. Melville Engle
J. Melville Engle

Dated: 9/15/2022

PREDICTIVE ONCOLOGY INC.

By: /s/ Bob Myers
Its: Chief Financial Officer

AGREEMENT AND RELEASE

This Agreement and Release ("Agreement") is made by and between J. Melville Engle ("Employee") and Predictive Oncology Inc., a Delaware corporation ("the Company"), each of whom enter into this Agreement intending to be legally bound.

1. **Background.** The Company and Employee state the following facts and incorporate them by reference into this Agreement.

    a. Employee and the Company entered into an Employment Agreement, effective March 19, 2021 (the "Employment Agreement"). A true and correct copy of the Employment Agreement is attached as Exhibit A to the Agreement.

    b. The Company adopted a 2021 Long-Term Incentive Plan, effective May 17, 2021 (the "LTIP"), pursuant to which Employee was granted a restricted stock unit award.

    c. Employee voluntarily resigned his employment with the Company effective October 31, 2022 (the "Separation Date"), pursuant to that certain Transition and Separation Agreement, dated September 15, 2022 (the "Separation Agreement"), in connection with Employee's retirement.

    d. In accordance with and subject to the terms of the Employment Agreement and the LTIP, Employee is entitled to certain separation benefits upon termination without cause.

    e. Notwithstanding that Employee voluntarily resigned his employment and was not terminated without cause, the Company agrees to provide Employee with the separation benefits in accordance with the Employment Agreement and the LTIP, and as set forth in this Agreement.

    f. Even if Employee does not enter into this Agreement, he will receive timely payment of his base salary through the Separation Date and accrued, unused vacation pay as of the Separation Date.

    g. The Company and Employee now agree as follows.

2. **Separation Benefits.** In exchange for Employee's waiver and release of claims set forth in Section 3 and other promises set forth in this Agreement, and provided that Employee (i) signs, dates, and returns this Agreement between November 1, 2022, and November 4, 2022, and (ii) does not rescind or revoke this Agreement within the time period described in Section 4, the Company agrees to provide Employee with the following "Separation Benefits" (identified below in this Section 2) to which Employee would not otherwise be entitled without signing this Agreement:

a. **Severance Pay.** The Company will pay Employee $524,400 (gross), less applicable federal, state and local withholdings and other legally required withholdings ("Severance Pay") which is 12 months of Employee's base salary. This Severance Pay will be paid in 24 equal installments in the amount of $21,850 each on successive paydays beginning on the first payday after 20 days have passed from the date Employee signs and returns this Agreement to the Company.

b. **Pro-Rata 2022 Bonus.** The Company will pay Employee $139,000.08 (gross), less applicable federal, state and local withholding and other legally required withholdings, as a pro-rata bonus for 2022, in 24 equal installments in the amount of $5,791.67 each on successive paydays beginning on the first payday after 20 days have passed from the date Employee signs and returns this Agreement to the Company.

c. **Vesting of RSUs**. Notwithstanding the terms and conditions of the LTIP, the Company will cause the LTIP to vest with respect to 300,000 RSUs on the date 20 days after the date Employee signs and returns this Agreement to the Company and will deliver to Employee the shares of the Company's common stock issuable upon such vesting as soon as administratively practicable following such vesting, subject to collection from Employee of any tax withholding required to be remitted as a result of such delivery.

3. **Waiver and Release of Claims.** In exchange for the Separation Benefits set forth in Section 2, Employee agrees to unconditionally waive and release any and all claims, complaints, causes of action, or demands of whatever kind which Employee has or may have against the Released Parties (as defined below) to the maximum extent permitted by applicable law up to the moment Employee signed this Agreement, including any claims, complaints, causes of action, or demands relating in any way to Employee's employment with the Company and Employee's separation from employment with the Company including, but not limited to, the following:

a. All claims for any alleged unlawful discrimination, harassment, failure to accommodate, retaliation, interference, reprisal arising, or other alleged unlawful practices under any federal, state, or local law, statute, ordinance, or regulation, including, without limitation, rights or claims of age discrimination, harassment, and retaliation under the federal Age Discrimination in Employment Act ("ADEA"), federal Older Workers Benefit Protection Act ("OWBPA"), Minnesota Human Rights Act ("MHRA"), the Colorado Anti-Discrimination Act ("CADA"), and claims of discrimination, harassment, failure to accommodate, and retaliation under the Family and Medical Leave Act, the Americans with Disabilities Act, Title VII of the Civil Rights Act of 1964, Equal Pay Act, MHRA, CADA, and the Minnesota Whistleblower Act;

b. All claims arising out of Employee's employment and Employee's separation from employment including, but not limited to, claims based on alleged wrongful discharge, breach of contract, breach of implied contract, failure to keep any promise, breach of a covenant of good faith and fair dealing, breach of fiduciary duty, defamation, infliction of emotional distress, fraud, misrepresentation, negligence, constructive discharge, assault, battery, false imprisonment, invasion of privacy, interference with contractual or business relationships, Employee's activities, if any, as a "whistleblower," and any violation of any other principle of common law;

c. All claims for any other alleged unlawful employment practices related to Employee's employment or Employee's separation from employment arising under any federal, state, or local law, statute, ordinance, or regulation including, without limitation, Sections 1981 and 1983 of the Civil Rights Act of 1866, the Employee Retirement Income Security Act, the Worker Adjustment and Retraining Notification Act, the Fair Credit Reporting Act, and the National Labor Relations Act;

d. All claims for any other form of pay, compensation, or employee benefits of any kind that is not provided in this Agreement including, without limitation, bonuses, commissions, deferred compensation, stock-based incentive compensation, restrict stock units, stock options, phantom stock, equity of any kind, vacation pay, expense reimbursement, and any other claims under any applicable federal, state, and local law, statute, ordinance, or regulation to the fullest extent permitted by law;

e. All claims Employee has now, whether or not Employee currently knows about or suspects the claims; and

f. All claims for attorneys' fees, costs, or interest.

Employee understands and agrees that the above list contains examples only and does not contain all claims that Employee is releasing. By signing this Agreement, Employee is fully and finally waiving and releasing, to the fullest extent permitted by law, all claims against the Released Parties. Employee agrees that the Company's payment of the Separation Benefits is full and fair payment for the waiver and release of Employee's claims and has a value greater than anything Employee is entitled to if Employee does not sign this Agreement. Notwithstanding anything set forth in this Agreement, specifically excluded from the waiver and release of claims set forth above are claims or disputes that: (i) by law cannot be released in a private agreement (such as workers' compensation claims); (ii) relate to any rights of indemnification afforded Employee by statute or by common law, including any insurance coverage maintained by or on behalf of the Company; (iii) arise after the date Employee signed this Agreement; or (iv) relate to the obligations of Employee or the Company under this Agreement.

For purposes of this Agreement, the term "Released Parties" means the Company and all of the Company's past and present parents, subsidiaries, and affiliated companies, and all and each of the past and present employees, officers, officials, managers, members, directors, agents, insurers, representatives, counsel, shareholders, owners, attorneys, partners, predecessors, successors, and assigns of any and all of the foregoing entities and persons. In addition, for purposes of Section 3, the term "Employee" means J. Melville Engle and any person who has or obtains any legal rights or claims against the Company or the Released Parties through J. Melville Engle.

4. **Employee's Legal Rights.**

a. **Advice to Consult With an Attorney.** This Agreement is a legal document. Employee has been advised in writing to consult with an attorney prior to executing the Agreement.

b. **Period to Consider this Agreement and Revoke this Agreement.** Employee has twenty-one (21) days from the date he receives this Agreement to consider the offer as expressed, including Employee's waiver and release of rights and claims of age discrimination under the ADEA and OWBPA, and decide whether to sign this Agreement. The Company has extended this time period to November 4, 2022. Signing this Agreement before November 4, 2022 constitutes a waiver by Employee of any remaining time period for review and consideration to which Employee may be entitled. Employee agrees that any changes to this Agreement, whether they are material or immaterial, do not restart the running of the 21-day consideration period. If Employee does not sign this Agreement on or before November 4, 2022, the offer contained within this Agreement will expire.

If Employee signs this Agreement, Employee will then be entitled to revoke this Agreement within seven (7) days after the date on which Employee signed this Agreement and the Agreement shall not become effective or enforceable until the revocation period has expired.

c. **Period to Rescind The Release of Claims.** Employee understands that Employee has the right to rescind Employee's waiver of discrimination and retaliation claims under the MHRA within fifteen (15) calendar days after the date on which Employee signs this Agreement. The 15-day rescission period and 7-day revocation period run at the same time, and this Agreement shall not become effective or enforceable until both periods have expired without Employee's rescinding or revoking this Agreement.

d. **Revocation/Rescission Procedure.** To revoke or rescind, Employee must put the revocation/rescission in writing and deliver it to the Company by hand to Bob Myers or mail within the revocation or rescission period. If Employee delivers the revocation/rescission by mail, it must be: (1) postmarked within the 7-day period to revoke his waiver of claims under the ADEA, and within the 15-day period to rescind his waiver of claims under the MHRA; (2) properly addressed to Predictive Oncology Inc. c/o Bob Myers, 2915 Commers Drive, Suite 900, Eagan, MN 55121, and (3) sent by certified mail, return receipt requested.

e. **Effect of Revocation/Rescission.** If Employee rescinds or revokes this Agreement as described in this Section 4, Employee understands that (i) this Agreement is null and void, (ii) the Company shall have no further obligation under this Agreement, (iii) Employee will not receive the Separation Benefits in Section 2 of this Agreement or any other benefits listed within this document, and (iv) Employee's employment will still end on the Separation Date.

5. **Filings.** Employee understands that, without being penalized or having an obligation to notify the Company, this Agreement does not prohibit Employee from filing an administrative charge of discrimination or complaint with the Equal Employment Opportunity Commission, National Labor Relations Board, Occupational Safety and Health Administration, Securities and Exchange Commission, Civil Rights Division, Minnesota Department of Human Rights, Colorado Civil Rights Commission, or any other federal, state, or local governmental agency or commission or law enforcement agency ("Government Agencies"). Employee understands that this Agreement does not limit Employee's ability to communicate with any Government Agencies or otherwise participate in any investigation or proceeding that may be conducted by any Government Agencies, including providing documents or other information, without notice to the Company. If Employee had filed or files a charge or complaint, Employee agrees that the Company's payment of the Separation Benefits completely satisfies any and all claims for monetary relief in connection with such charge or complaint, except that this Agreement does not limit Employee's right to receive an award for information (1) provided pursuant to the Securities and Exchange Commission's whistleblower protections and incentives; or (2) provided to any other Government Agencies. Employee is not entitled to any other monetary relief of any kind with respect to the claims that Employee has released in this Agreement unless Employee's waiver and release of claims is deemed unlawful or otherwise invalid.

6. **Governing Law/Venue.** The laws of the State of Minnesota will govern the validity, construction, and performance of this Agreement, without regard to the conflict of law provisions of any other jurisdictions. Employee irrevocably consents to the exclusive jurisdiction of courts in Minnesota for the purposes of any action arising out of or related to this Agreement or any dispute between the Company and Employee, including any actions for temporary, preliminary, and permanent equitable relief. Employee irrevocably waives Employee's right, if any, to have any disputes between the Company and Employee arising out of or related to this Agreement decided in any jurisdiction or venue other than a state or federal court in the State of Minnesota.

7. **Additional Agreements and Understandings.**

a. **The Company Property.** Employee has returned to the Company all the Company property in Employee's possession or under Employee's control including, but not limited to, all corporate credit cards, identification badges, computer hardware and software, cell phones, tablets, PDAs, books, records, documents, data, access cards, financial data, confidential information, trade secrets, files, notebooks, passwords, plans, sales reports, records, and all other property, equipment, or information owned by the Company or to which Employee was provided access by the Company during Employee's employment. By signing this Agreement, Employee represents that Employee has returned all the Company property and that Employee no longer possesses or has access to the Company property. Notwithstanding the foregoing, the Company agrees that Employee may keep the Company-owned computer in his possession, and by signing this Agreement, Employee represents that there is no Company information maintained or stored on the computer.

b. **Employment Agreement.** The Employment Agreement contains valid and enforceable restrictions on Employee's competition with the Company and solicitation of others, both during and after employment with the Company. Employee acknowledges and agrees that the Employment Agreement is fully enforceable and survives the termination of Employee's employment.

The Company advises Employee as follows under the federal Defend Trade Secrets Act: An individual shall not be held criminally or civilly liable under any Federal or State trade secret law for the disclosure of a trade secret that — (A) is made — (i) in confidence to a Federal, State, or local government official, either directly or indirectly, or to an attorney; and (ii) solely for the purpose of reporting or investigating a suspected violation of law; or (B) is made in a complaint or other document filed in a lawsuit or other proceeding, if such filing is made under seal. In addition, an individual who files a lawsuit for retaliation by an employer for reporting a suspected violation of law may disclose the trade secret to the attorney of the individual and use the trade secret information in the court proceeding, if the individual — (A) files any document containing the trade secret under seal; and (B) does not disclose the trade secret, except pursuant to court order. Nothing in this Agreement is intended to conflict with 18 U.S.C. § 1833(b) or create liability for disclosures of trade secrets that are expressly allowed by 18 U.S.C. § 1833(b).

c. **Transition and Cooperation.** The Employee will cooperate with the Company and use his best efforts to be available, on a reasonable basis, to discuss or address issues or questions that may arise after the Separation Date relating to Employee's employment and position for the purpose of achieving a smooth transition of Employee's former job duties and responsibilities. Employee also agrees to be available to and cooperate with the Company and its counsel in connection with any investigation, administrative proceeding or litigation relating to any matter, occurring during Employee's employment, in which he was involved or of which he has knowledge. Employee understands and agrees that such cooperation includes, but is not limited to, making himself available to the Company and/or its counsel upon reasonable notice for: interviews and factual investigations; appearing to give testimony without requiring service of a subpoena or other legal process; volunteering to the Company or its counsel pertinent information; and turning over all relevant documents which are or may come into his possession.

d. **Consideration.** Employee agrees that (i) the Separation Benefits in Section 2 are above and beyond that to which Employee would be entitled if Employee did not sign this Agreement, (ii) the Separation Benefits in Section 2 constitute independent and sufficient consideration for all aspects of this Agreement, and (iii) Employee is not eligible for any other payments or benefits except for those expressly described in this Agreement, provided that Employee signs and returns this Agreement within the specified time period and does not rescind or revoke this Agreement.

e. **Non-Disparagement.** Employee agrees not to make disparaging or defamatory remarks about the Company or the Company's services, products, or other matters pertaining to its business. The Company, on behalf of itself and its management team, agrees not to make disparaging or defamatory remarks about Employee or Employee's services or other matters pertaining to Employee. This non-disparagement provision does not apply to legally protected communications and does not prohibit Employee or the Company from filing an administrative charge or complaint with, or cooperating, assisting, testifying, or participating in an investigation or legal proceeding conducted or initiated by, any Government Agencies.

f. **Non-Admission.** It is expressly understood that this Agreement does not constitute, nor shall it be construed as, an admission by the Company of any liability or unlawful conduct whatsoever. The Company specifically denies any liability or unlawful conduct on the Company's part.

g. **Successors and Assigns.** This Agreement is personal to Employee and may not be assigned by Employee without the written agreement of the Company. The rights and obligations of this Agreement shall inure to the successors and assigns of the Company.

h. **Severability.** If a court finds any term of this Agreement to be invalid, unenforceable, or void, Employee and the Company agree that the court shall modify such term to make it enforceable to the maximum extent possible. If the term cannot be modified, Employee and the Company agree that the term shall be severed and all other terms of this Agreement shall remain in effect. Employee and the Company agree that Employee's waiver and release of claims should be interpreted as broadly as possible to achieve Employee's intention of releasing all claims against the Released Parties.

i. **Entire Agreement.** This Agreement constitutes the sole understanding of Employee and the Company with respect to the matters provided for herein. Employee and the Company agree that this Agreement supersedes and terminates any and all other written and oral agreements and understandings between Employee and the Company concerning separation benefits Employee may have been eligible for or entitled to from the Company. Notwithstanding anything in this Agreement to the contrary, Employee agrees and acknowledges that the Employment Agreement remains in full force and effect after the Separation Date. This Agreement may not be modified, altered, or changed in any way except by written agreement signed by Employee and an authorized representative of the Company.

j. **No Waiver.** No claim or right arising out of a breach or default under this Agreement may be discharged by a waiver of that claim or right unless the waiver is made in writing and signed by an authorized representative of the Company. A waiver by any party of a breach or default of the other party of any provision contained in this Agreement shall not be deemed a waiver of future compliance of such provisions, and such provisions shall remain in full force and effect.

k. **Remuneration.** Employee acknowledges and agrees that the Company will pay Employee any and all monies, wages, salary, accrued and unused paid time off, expenses, bonuses, and commissions (if applicable) due to Employee through the Separation Date. Employee is not entitled to any additional remuneration from the Company other than the consideration outlined within this Agreement. In addition, Employee acknowledges that Employee is not aware of any time worked during Employee's employment for which Employee has not already been fully compensated.

l. **Acknowledgements.** Employee acknowledges and agrees that: (i) Employee has not suffered any work-related injury for which Employee has not already filed a claim; and (ii) Employee has been properly provided any leave of absence including for Employee's own or a family member's health condition.

m. **Taxes.** Employee acknowledges that Employee has not relied on any tax advice provided by the Company and that, if necessary, Employee is solely responsible for properly reporting the payment received pursuant to this Agreement and paying any applicable taxes, penalties, and interest. Employee acknowledges and agrees that Employee has been provided with the opportunity to consult legal and financial counsel with respect to the tax treatment of the Separation Benefits Employee will receive pursuant to this Agreement and on account of Employee's separation from employment. Employee has been advised by the Company to consult with such counsel.

n. **Accepting/Signing this Agreement.** Employee agrees not to sign this Agreement prior to the end of Employee's work day on the Separation Date.

[*Signature page follows*]

Dated: _____

_____
J. Melville Engle

Dated: _____

PREDICTIVE ONCOLOGY INC.

By: _____
Its: _____